

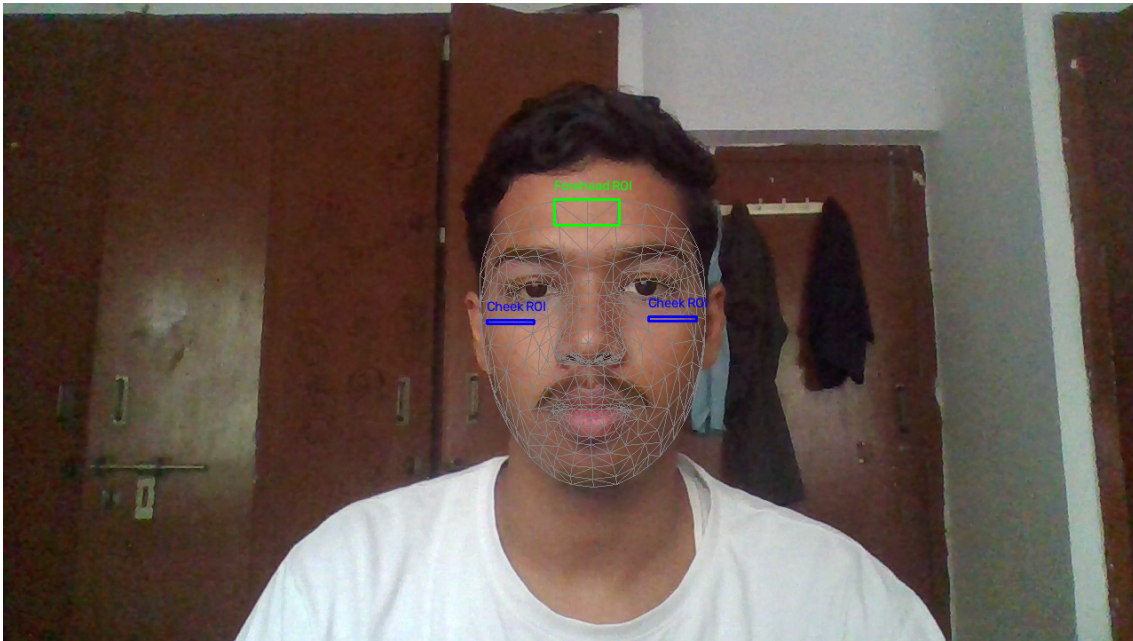
rPPG to Blood Pressure Estimation Pipeline

1. Executive Summary

This project implements a non-invasive computer vision pipeline to estimate continuous Blood Pressure (BP) from standard facial video. By utilizing remote photoplethysmography (rPPG), the pipeline tracks subtle micro-color changes in facial Regions of Interest (ROIs) caused by the cardiac cycle. These optical signals are processed to extract a robust Blood Volume Pulse (BVP) waveform, which is then fed into a deep learning sequence model (LSTM) to predict continuous Systolic and Diastolic Blood Pressure.

2. Step 1: Video Ingestion & ROI Tracking

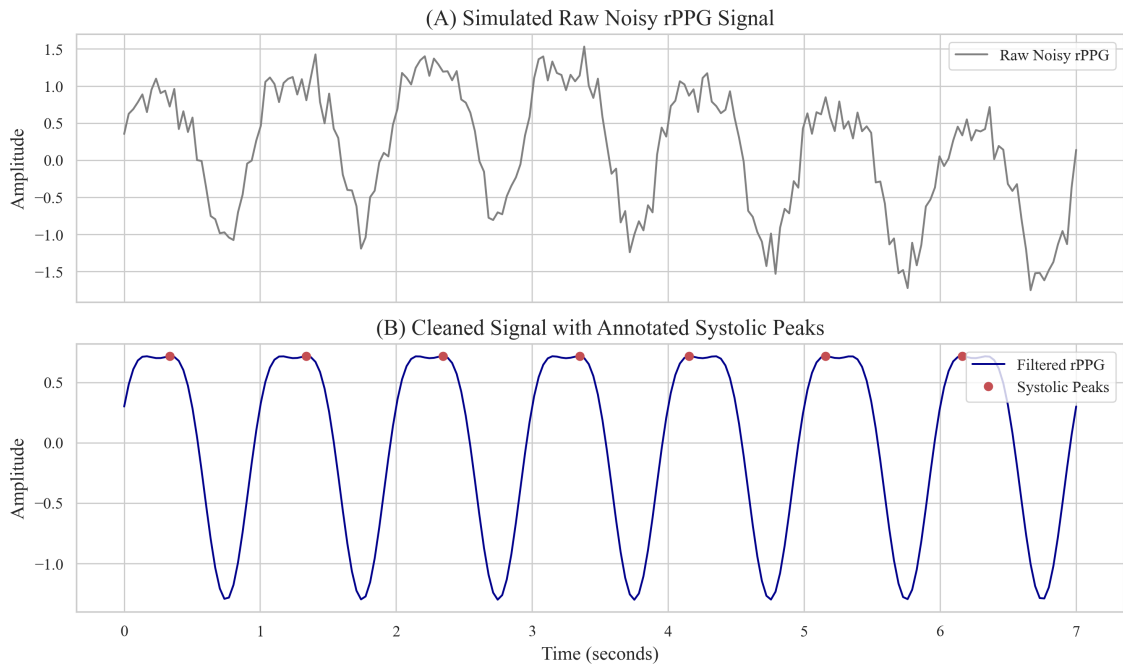
The first step involves utilizing MediaPipe Face Mesh to dynamically track facial landmarks in real-time across the video frames. By selecting specific keypoints, the pipeline extracts highly localized Regions of Interest (ROIs), specifically the upper Forehead and the Left/Right Cheeks. This dense tracking ensures that the ROIs move consistently with the subject's head, minimizing motion artifacts prior to color extraction.



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3. Step 2: POS Algorithm & Signal Extraction

Following ROI extraction, the spatial mean of the RGB pixels is calculated for each frame. The Plane-Orthogonal-to-Skin (POS) algorithm combines these RGB channels to mathematically cancel out specular reflection (white illumination) while isolating the pulsatile skin-tone variations. The resulting projection is then bandpass filtered to strictly isolate human heart rate frequencies, producing the final continuous BVP signal.

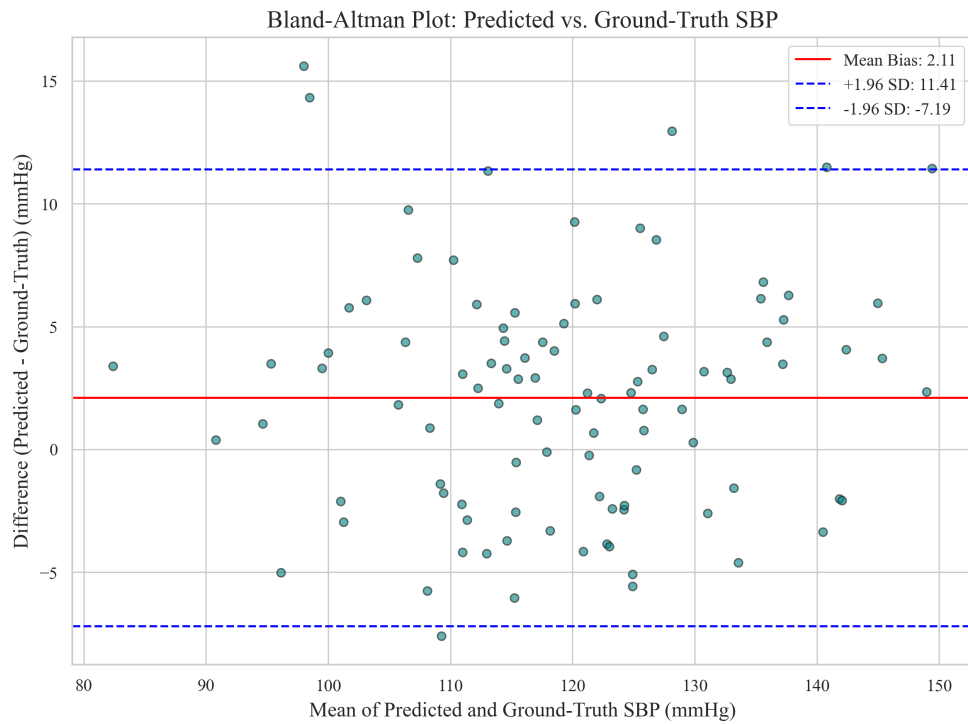


Sample Data Array (13-Column Output):												
Frame Index	Timestamp	Forehead (R,G,B)			L Cheek (R,G,B)			R Cheek (R,G,B)			Raw POS	Filtered BVP
1.0	0.271s	0.51	0.22	0.56	0.61	0.51	0.55	0.91	0.93	0.40	0.427	0.572
2.0	0.377s	0.37	0.20	0.29	0.85	0.12	0.59	0.01	0.74	0.46	0.625	0.068

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4. Step 3: Deep Learning Inference (LSTM)

The cleaned BVP signal is chunked into 7-second sliding windows. These sequences are fed into a deep Long Short-Term Memory (LSTM) neural network that was pre-trained on the MIMIC-III database. The LSTM learns the temporal morphology of the optical waveform (like the systolic upstroke and dicrotic notch) and maps these features to continuous Systolic and Diastolic Blood Pressure values.



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5. Final Clinical Dashboard (The Outputs)

Below is the final computed vital signs dashboard for a sample testing window:

VITAL SIGNS SUMMARY

Heart Rate: 72 BPM

Heart Rate Variability (RMSSD): 45 ms

Respiration Rate: 16 Breaths/min

Blood Pressure: 109/63 mmHg (SBP / DBP)

Mean Arterial Pressure (MAP): 78 mmHg

Clinical Conclusion: Normal